

6.4.7 Spike sensitivity

Single voltage spikes demonstrate why local and global metrics must be considered together. Full-window ΔMAE remains below 4×10^{-4} SOC for every estimator and is effectively zero for DM, HDM, and DD. Event-aligned analysis nevertheless identifies a reproducible DD transient. Its mean spike-sample penalty is approximately 7.7×10^{-3} SOC across cells, compared with near-zero direct penalties for the other representatives. Figures 6.12 and 6.13 show that this response varies across cells and SOH states and decays after the event.

DM and HDM do not use voltage as a continuous SOC-correction signal, so an isolated spike away from the full-charge threshold has little direct effect. HECM processes voltage through its observer residual, which filters one inconsistent measurement through the physical model and covariance weighting. DD receives the impulse simultaneously through absolute voltage and dU/dt , and both disturbed values enter its recurrent input context. This provides a plausible mechanism for the larger but short-lived response. A channel-removal ablation would be required to isolate the individual contribution of each voltage feature.

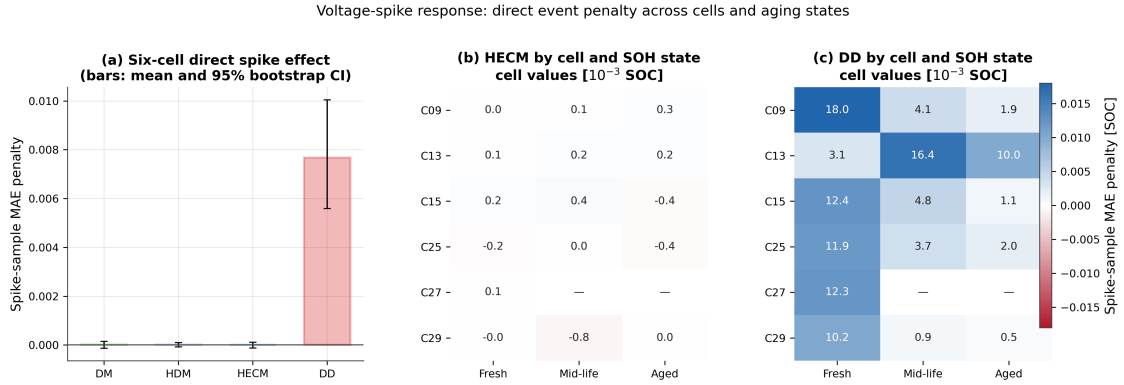


Figure 6.12: Voltage-spike sensitivity across six cells. Panel (a) reports the mean event-sample MAE penalty with a hierarchical 95% confidence interval. Panels (b) and (c) resolve HECM and DD penalties by cell and SOH state. The event-aligned metric exposes the local DD response that is obscured by full-window ΔMAE .